Keaton Van Patten

AI Engineer / Forward Deployed Engineer

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SUMMARY

Builder-operator who ships production AI systems end to end, deploys them into environments he does not own, and measures what they actually do. Sole developer of a scheduling system a pediatric therapy clinic has used daily since August 2026, specified from requirements taken face to face with clinic staff. Also builds and operates a grounded retrieval service over 3,190 files behind a release-blocking evaluation harness, and a 122-tool agent under fail-closed tool authorization. Every claim here traces to a number measured on real hardware, including the ones that proved earlier claims wrong.

I build through AI-assisted (generative) development: I design, direct, test and operate the systems, and the code is produced with AI tooling under my direction.

CORE SKILLS

AI/LLM agent orchestration, multi-agent systems, RAG (BM25 · vector embeddings · RRF hybrid search), tool calling / function calling, MCP (Model Context Protocol), prompt engineering, LLM evaluation harnesses, LLM-as-judge, guardrails & prompt-injection defense, human-in-the-loop gating, cost attribution & model routing, local inference (Ollama · CUDA · ROCm), Claude API, OpenAI-compatible APIs, DeepSeek, Whisper STT, TTS

Engineering Python, FastAPI, TypeScript/JavaScript, SQL/SQLite/PostgreSQL, Svelte, Electron, REST API design, WebSockets/SSE, PHP, Docker, CI/CD (GitHub Actions), pytest, Fly.io, Tailscale, Windows/Linux administration

Cloud & identity Microsoft 365 / Entra ID / Azure administration (5+ years), AWS (labs and hands-on study, not production)

Customer-facing requirements discovery with non-technical users, stakeholder management, technical training, incident communication, security-review facilitation, vendor evaluation & management

SELECTED PRODUCTION SYSTEMS

Clinic Scheduling System + Outage Failover

pediatric therapy clinic, live since August 2026

In daily use by clinic staff: 573 audited actions in the first 18 days.

- Sole developer of a workload-based scheduling system replacing a fully paper process: 30 REST endpoints and 15 role-gated pages in PHP, SQLite and a zero-dependency JavaScript PWA.
- Designed an 8-component weekly workload model computing therapist utilization to the quarter-hour with hard over-capacity blocking, plus a filter-and-rank substitute selector for absence coverage.
- After a real 30-minute production outage, built and deployed a zero-dependency Node failover mailer on Fly.io in under 48 hours: outbound-only with no public IP, and PHI minimized server-side so only de-identified case codes leave the host.
- Commissioned a pre-launch security audit through my own AI security-review desk and closed both launch-blocking criticals it found. Shipped Argon2id auth, email OTP two-factor sign-in, CSRF-before-body-parse hardening, strict CSP/HSTS and a 14-action audit log.

RAG Reference Service ("Librarian")

grounded retrieval with a release-blocking eval harness

- Built and operate a grounded question-answering service over 3,190 files: SQLite full-text Okapi BM25 alongside a dense cosine arm fused by Reciprocal Rank Fusion, a custom rerank stage, and graceful degradation to keyword-only.
- Served to AI coding agents as an MCP server behind a multi-provider LLM fallback chain, most answers returned by the low-cost primary: 613 live queries in 30 days, mostly machine to machine.
- Gated releases on an evaluation harness of 22 graded questions, the deterministic ones required to pass with every LLM provider offline. Measured accuracy went from 86.4% to 100%.

Multi-Provider LLM Platform

cost governance and routing

- Operate a metered multi-provider model fleet under a USD-exact per-call ledger: 2,129 calls over 61 days, every row carrying the use-case surface it served and what it cost.
- Cost per model call fell from $0.048 to $0.014 while volume grew, by profiling provider pricing, latency and data sensitivity, then re-routing background work to cheaper tiers and pinning latency-critical and private-data paths to local GPUs. A routing-policy outcome, not a controlled A/B.
- Stood up a self-hosted LiteLLM control plane on PostgreSQL issuing per-use-case virtual API keys with hard 30-day budgets, with a raw-socket TCP probe driving automatic gateway-to-direct fallback. Rejected Langfuse tracing after an audit showed dozens of call paths bypass any proxy.

Local-First Agent Platform ("Companion Hub")

122-tool production agent

- Architected a local-first agentic platform in Python, FastAPI, Svelte and Electron whose agent dispatches 122 real tools, under a three-layer fail-closed authorization model: per-surface allowlists, chokepoint denylists and a credential-aware filesystem perimeter, with dedicated gate tests.
- Wrote a token-budgeted prompt architecture that holds the static prefix byte-identical across turns for inference-cache hits, with golden-file regression tests, and cut small-model tool-catalog overhead from 7.4K to 1.4K tokens.
- Chose the local security-triage model adversarially: it resisted every prompt injection in the benchmark, and a candidate was rejected for obeying injected verdicts. The review desk it powers has run 110 consultations across 22 projects.

Autonomous Incident Desk

three-tier deflection ladder

- Runs a self-healing service desk watching production backends on a fifteen-second probe loop: 612 incidents handled, 72.9% auto-resolved at a sub-minute mean time-to-resolution, almost none escalated to a human.
- Tiered the cost of deflection: scripted restart playbooks first, then LLM triage on only the 3.3% of tickets they miss, diagnosis-only by contract, with a kill-regex validator, a restart circuit breaker and an independent kill-witness auditor bounding blast radius. These are infrastructure incidents, not support tickets.

EXPERIENCE

HRBC Inc., Systems Manager & Software Developer

Sacramento, CA · Sept 2020 to Present

- Sole technical owner for a mortgage brokerage: systems administration, Microsoft 365 and Entra ID, user access management, vendor systems (LOS, CRM, pricing engines), incident response, and training for loan officers. Over five years of requirements discovery run directly with non-technical stakeholders.
- Delivered an internal operations hub with an embedded AI assistant: a custom agentic tool-use loop written directly against the Anthropic API with no agent framework, 8 typed JSON-schema tools, a bounded round-trip budget and mutation tracking, over a 15-system vendor registry that stores credential locations and never secrets. Backend complete and demonstrated via API; the UI was never wired up.
- Licensed mortgage loan originator with an active NMLS license: advised clients directly, trained and supervised loan officers and assistants.

Pediatric therapy clinic, Software Developer, independent engagement

June 2026 to Present

- Sole developer of the clinic's scheduling system and its outage failover mailer, with requirements gathering, release authorization, security remediation and incident communication owned end to end.

MP NexLevel, Utility Technician

2018 to 2020

- Crew lead on utility infrastructure projects; equipment operation and maintenance.

EDUCATION & CERTIFICATES

Google IT Support Professional Certificate · Google Computer Networking Certificate